Recursion, 2hrs ✓ | ~~storage class~~, 2hrs ✓ | ~~Pointer-Basics~~, 4hrs

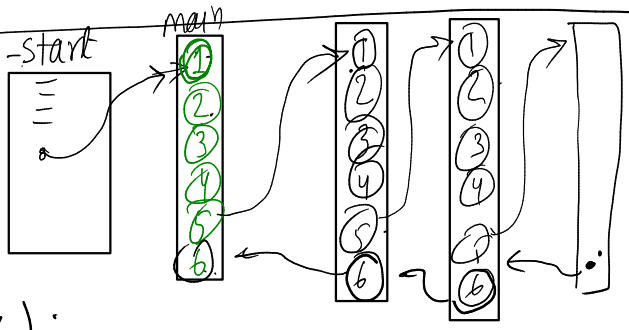
What's Recursion / Recursive function?

- A function calling itself is recursive function.
- calling function and called function same.

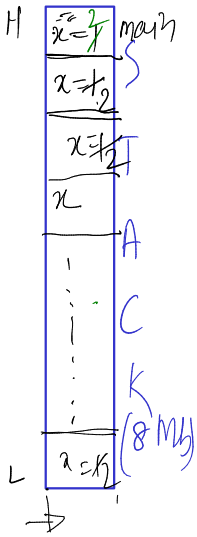
loop vs recursion

Why Recursion?

- For Repeatation of code




```
main() {
1) int x = 1;
2) pf("\n in main x = %d \n", x);
3) ++x;
4) if(x<5)
5)     main();
6) pf("Bye from main \n");
}
```



stack overflow results in segmentation fault.

Segmentation fault

→ A process has dedicated memory area to access. If the process-
↳ (text, data, stack)

trying to access memory area, beyond its allocation, then exception/Interrupt occurs.

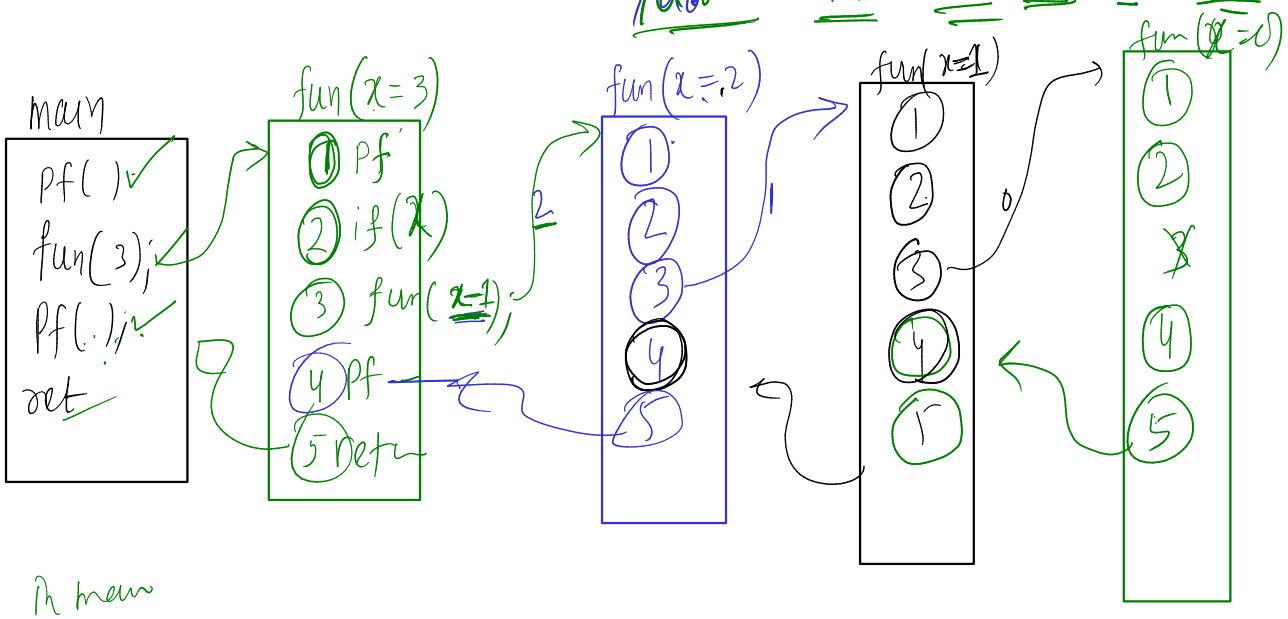
This unauthorised-memory-access interrupt, results in process termination

$ ./a.out ↵

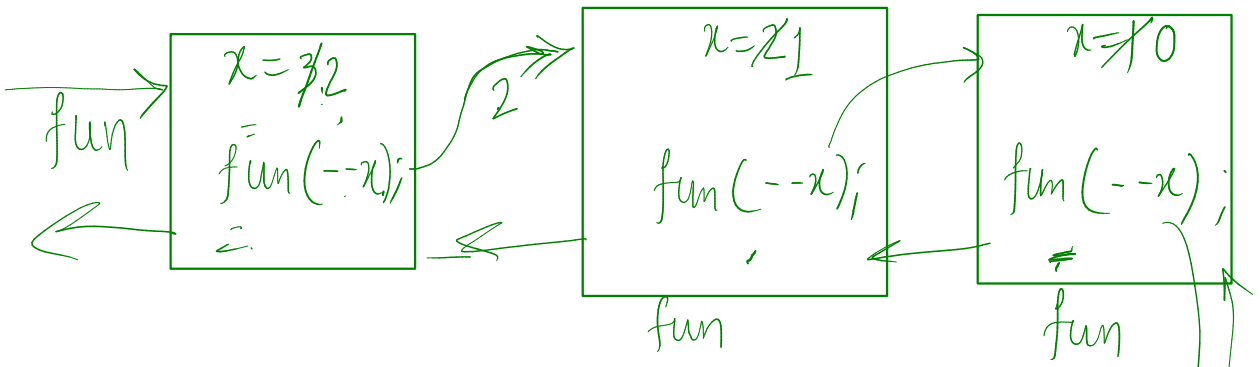
```
main( int argc){
  1
  3
  6
}
```

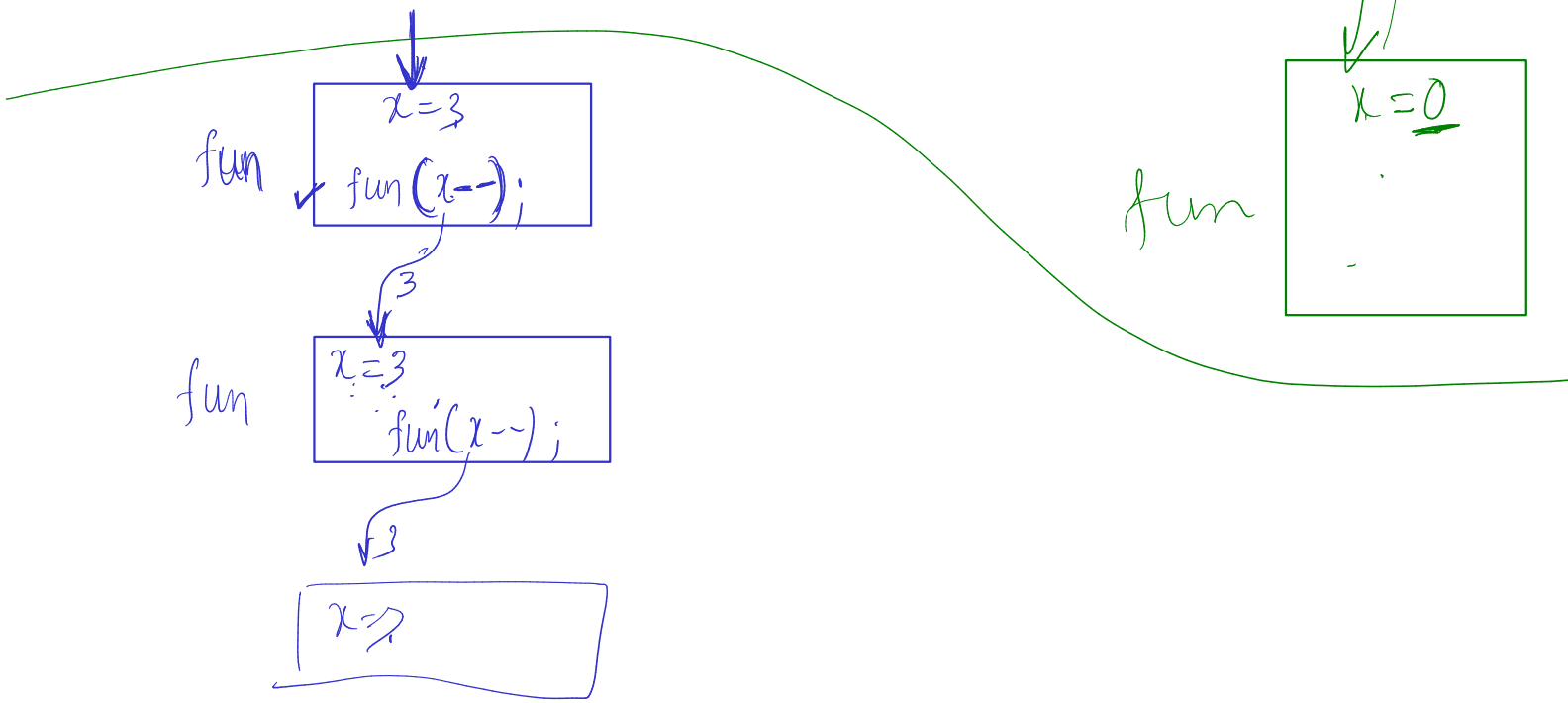
./a.out 12 34

./a.out xyz abc 11 22 33



In main
x = 3
Hi x = 2
Hi x = 1
Hi x = 0
Bye x = 0
Bye x = 1
Bye x = 2
Bye x = 3.





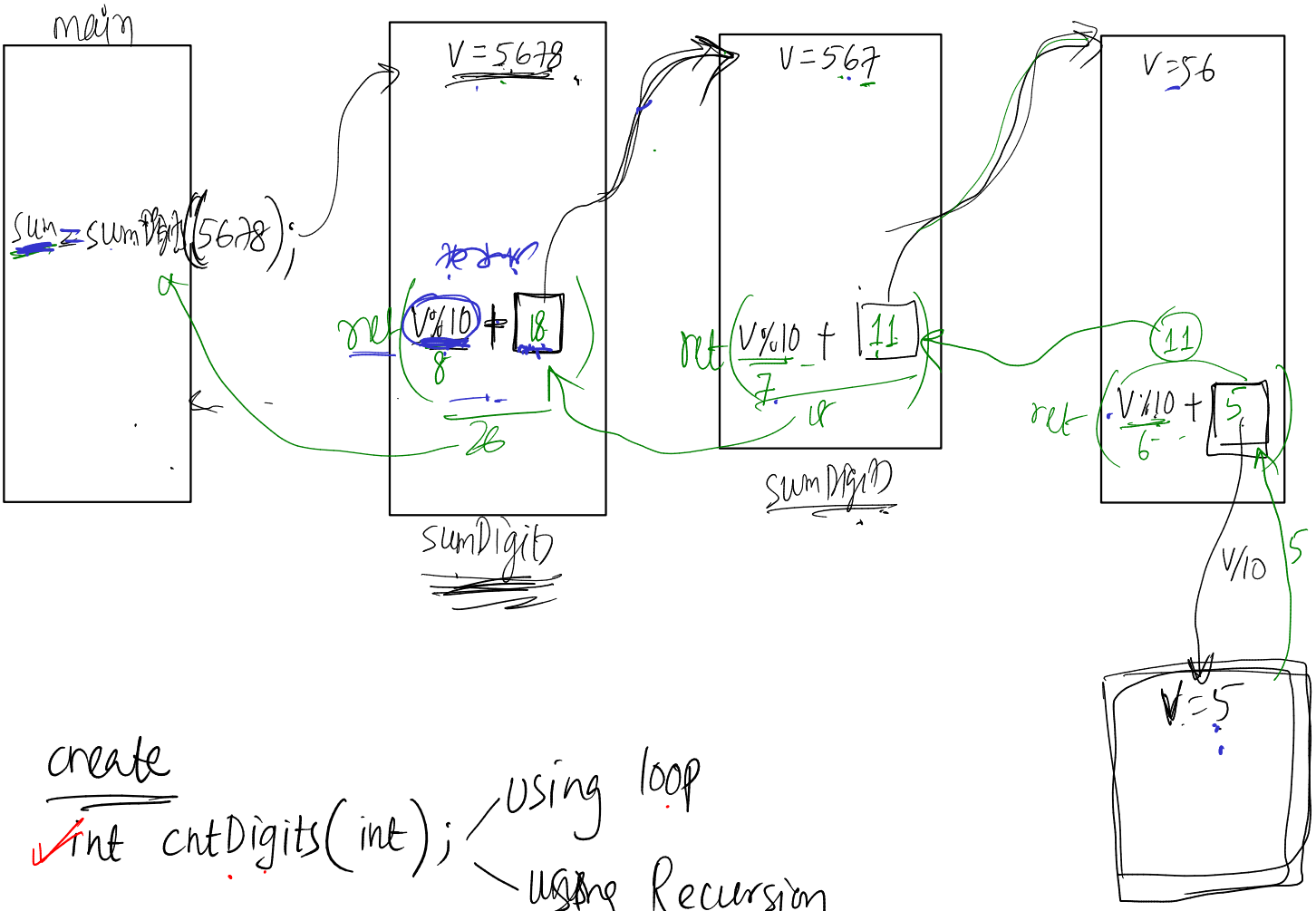
Recursion Based Design

```
main() {
    scf("%d", &data);
sum = sumDigits(data);
}
```

```
int sumDigits(int v)
{
  if (v/10==0)
    return v;
  return (v%10 + sumDigits(v/10));
}
```

```
int sumDigits(int var)
{ //loop based
  int s = 0;
  while(var) {
    s = s + var%10;
    var /= 10;
  }
  return s;
}
```

s	s = s + D
0	0+7
7	7+6
13	13+0
13	13+5
18	



create

int cntDigits(int); ⟨ using loop / using Recursion

create

int factorial(int); ⟨ using loop / using recursion

Tomorrow (sunday) : 10:30 to 12:30

↳ loop Vs Recur